Comparative Analysis of Phenotypic Antimicrobial Resistance Patterns in Escherichia coli and Klebsiella pneumoniae Using Computational Methods

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Abstract

Antimicrobial resistance (AMR) poses a major threat to global public health, particularly in clinically significant Gram-negative pathogens such as *Escherichia coli* and *Klebsiella pneumoniae*. These organisms are common causes of hospital- and community-acquired infections and are increasingly associated with multidrug resistance. In this study, publicly available phenotypic antimicrobial susceptibility data were analyzed to compare resistance patterns between *E. coli* and *K. pneumoniae*. Over 80,000 susceptibility records were examined across multiple antibiotic classes. Exploratory data analysis was performed to assess organism-level and antibiotic-specific resistance trends, followed by basic machine learning–assisted feature analysis to support resistance pattern differentiation. The results demonstrate a substantially higher proportion of resistant phenotypes in *K. pneumoniae* compared to *E. coli*, with notable resistance observed against β-lactams and fluoroquinolones. This project highlights the value of integrating computational approaches with clinical microbiology data to better understand resistance trends relevant to antimicrobial stewardship and future biomedical research.

Background and Rationale

Antimicrobial resistance has emerged as one of the most urgent challenges in modern medicine, limiting effective treatment options for common bacterial infections. Among Gram-negative pathogens, *Escherichia coli* and *Klebsiella pneumoniae* are of particular concern due to their high prevalence in urinary tract infections, bloodstream infections, and hospital-acquired pneumonia. These organisms are frequently exposed to broad-spectrum antibiotics, accelerating the development and spread of resistance mechanisms such as extended-spectrum β-lactamases and carbapenemases.

In routine clinical microbiology laboratories, antimicrobial susceptibility testing is performed on individual patient isolates to guide therapy. However, aggregated susceptibility data across large datasets can provide deeper insights into resistance trends, organism-specific behavior, and antibiotic performance. Such analyses are essential for informing antimicrobial stewardship programs and identifying emerging resistance threats.

While laboratory-generated susceptibility results are abundant, their systematic analysis using computational tools remains underutilized in many clinical and academic settings, particularly in low- and middle-income countries. This project was designed to bridge clinical laboratory knowledge with computational analysis by examining real-world phenotypic AMR data and exploring whether resistance patterns can be meaningfully differentiated between *E. coli* and *K. pneumoniae* using data-driven approaches.

Methods

Phenotypic antimicrobial resistance data were obtained from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC, formerly PATRIC), a publicly curated database widely used in microbial genomics and AMR research. Separate datasets for *Escherichia coli* and *Klebsiella pneumoniae* were downloaded in CSV format and analyzed using Python.

Each dataset contained isolate-level information, including the tested antibiotic, resistance phenotype (susceptible, intermediate, or resistant), laboratory testing method, and testing standards. An organism label was added to each dataset prior to merging to enable comparative analysis. Data cleaning steps included removal of incomplete records for selected analyses and harmonization of categorical variables.

Exploratory data analysis was performed to assess overall resistance distributions, organism-specific resistance proportions, and antibiotic-level resistance patterns. Crosstabulation and normalization techniques were used to compare resistance frequencies across organisms. For computational analysis, resistance phenotypes were simplified to binary outcomes (resistant vs. susceptible), and selected categorical features were encoded for preliminary machine learning–assisted evaluation using logistic regression. All analyses were conducted using pandas, NumPy, matplotlib, seaborn, and scikit-learn. This analysis was conducted for exploratory and comparative purposes rather than predictive clinical deployment.

Key Findings and Significance

Comparative analysis revealed marked differences in antimicrobial resistance profiles between the two organisms. *Klebsiella pneumoniae* exhibited a substantially higher proportion of resistant phenotypes compared to *Escherichia coli* across the analyzed dataset. In particular, resistance to commonly prescribed antibiotics such as ampicillin, ceftriaxone, and fluoroquinolones was notably higher in *K. pneumoniae*, consistent with its known capacity to acquire and disseminate resistance determinants.

In contrast, both organisms demonstrated relatively preserved susceptibility to certain aminoglycosides and carbapenems, including amikacin and meropenem, highlighting their continued clinical relevance for severe infections. Machine learning–assisted feature analysis supported the observation that organism identity and antibiotic class were strong contributors to resistance pattern differentiation.

Although this study was exploratory in nature, it demonstrates how clinically generated phenotypic data can be leveraged using computational tools to extract biologically meaningful insights. The project underscores the potential of integrating laboratory medicine with data-driven analysis and reflects preparedness for further training in microbiology, immunology, and translational biomedical research within data-informed laboratory environments.